2025 AMC 10B Problem 17 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10B Problem 17. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10B Problem 17

Problem:

Consider a decreasing sequence of n positive integers $x_1 > x_2 > \cdots > x_n$ that satisfies the following conditions:

- The average of the first 3 terms in the sequence is 2025.
- For all $4 \leq k \leq n$, the average of the first k terms is 1 less than the average of the first $k-1$ terms.

What is the greatest possible value of n?

Answer Choices:

A. 1013

B. 1014

C. 1016

D. 2016

E. 2025

Join the Discussion

Stuck on this problem or want to share your approach?

Continue the conversation and see what others are thinking: [View Forum Thread](https://wiki.randommath.com/amc10/2025/part-b/problem-17)

Solution:

From the second condition, we have that the average of the first 4 terms is 2024, the average of the first 5 terms is 2023... so inductively the average of the first k terms is $2028 - k$. This also implies that the sum of the first k terms is $k(2028 - k)$.

If we know the partial sums of a sequence, we may recover the terms by subtracting two adjacent terms. In other words, the kth term of this sequence is the sum of the first k terms minus the sum of the first $k - 1$ terms. This gives us:

$$x_k = k(2028 - k) - (k - 1)(2028 - (k - 1))$$

$$= (2028k - k^2) - (-2029 + 2030k - k^2) = 2029 - 2k$$

As the terms must be positive: $x_k > 0 \implies 2029 > 2k \implies k < 1014.5$ gives us the maximum possible value of n as $\boxed{\textbf{(B) } 1014}$.

Additionally, note that since $a_4 = 2021$, it is possible to assign values to a_1, a_2, and a_3 which does not affect the bound later down the line.